# Module 1: SRS & PRD Document

## Common Login Page & Student Dashboard Specification

**Module Name:** Student Portal & Authentication
**Document Type:** Combined SRS & PRD
**Target Platform:** Responsive Web (React.js)
**Version:** v2.1 Final

## 1. EXECUTIVE SUMMARY & SCOPE

This document specifies the software requirements and product definitions for the **Common Authentication Portal** and the **Student Dashboard**. It details authentication security, role redirection logic, the strict 10-minute check-in cutoff, campus geofenced auto-checkout, complaint submission, and live map tracking.

## 2. COMMON LOGIN PAGE SPECIFICATION

The system implements a unified entry point (`/login`) for all registered roles (Student, Teacher, Driver, Admin).

| Field / Component | Validation & Rules | Error Response Payload |
|---|---|---|
| **Username / User ID** | Required; String (USN for students, Employee ID for staff). | `"Invalid User ID format."` |
| **Password** | Required; Minimum 8 characters with bcrypt hash verification. | `"Invalid credentials provided."` |
| **Role Redirection Router** | Decodes JWT claim `role` post-authentication and routes dynamically. | Redirects to `/student/dashboard`, `/driver/dashboard`, `/teacher/dashboard`, or `/admin/dashboard`. |

```
[ User Submits Credentials ] | v [ Auth API Endpoint POST /api/auth/login ] | +---> Verify Password Ha
```

## 3. STUDENT DASHBOARD FUNCTIONAL REQUIREMENTS

### 3.1 Student Information Panel

Displays authenticated student record retrieved from the database: Student Name, USN, Class, Department, Phone, Email, Home Address, and assigned Stop Name.

### 3.2 Check-In Protocol & 10-Minute Cutoff Rule

- **Check-in Deadline:** Students must explicitly confirm boarding intention at least 10 minutes prior to their assigned stop's scheduled arrival time.
- **Cutoff Enforcement:** System rejects check-in attempts at or after `(scheduled_stop_time - 10 mins)`. Status remains `NOT CHECKED IN`.
- **Visual Indicators:** Displays `CHECKED IN` (Green) when confirmed, or `NOT CHECKED IN` (Red).

## 4. GEOFENCED AUTO-CHECKOUT & LIVE TRACKING

### 4.1 Campus Geofenced Auto-Checkout

To automate attendance recording, a circular GPS geofence (Radius: 200 meters) is established around the college campus center point.

```
[ Bus Coordinates Ingested ] ---> Check Distance to Campus Center (Lat: 12.9716, Lon: 77.5946) | Is Di
```

### 4.2 Live Bus Tracking Interface

- **100-Meter Position Updates:** Map marker refreshes position only when the bus has traveled $\ge 100\text{ meters}$ from the previously published coordinate.
- **Data Overlay:** Displays current bus marker icon, assigned driver details (Name, Phone), Bus Registration Number, Route line, and timestamp of last location update.
- **Stale Location Warning:** If no valid GPS reading is received for $> 2\text{ minutes}$, the interface displays a `LOCATION STALE` alert.

## 5. COMPLAINT SUBMISSION MODULE

Students can lodge complaints regarding driver behavior, timing delays, or vehicle condition directly to the Admin.

| Field Name | Type | Validation Rules |
|---|---|---|
| `complaint_id` | UUID | Auto-generated primary key. |
| `description` | Text | Mandatory; Non-empty string, min 10 characters. |
| `submitted_at` | Timestamp | Auto-captured UTC server timestamp. |
| `status` | Enum | Defaults to `OPEN`. Shifts to `RESOLVED` when Admin updates query. |

## 6. NON-FUNCTIONAL & SECURITY REQUIREMENTS

- **Security:** All API communication enforced over HTTPS/TLS 1.3. JWT tokens stored in HTTP-only cookies.
- **Data Privacy:** Students can only view their own profile and assigned bus telemetry. Cross-student data exposure is strictly prohibited.
- **Performance:** Dashboard asset loading target $< 1.0\text{s}$ over 4G network. Map rendering under $200\text{ms}$.